Chapter 16

Mediastinum and chest wall

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Mediastinal masses

Key facts

- Most mediastinal masses in children and young adults are malignant.
- Anterior mediastinal masses should be removed, unless lymphoma is likely, which is better treated with adjuvant therapy.
- ***Thymoma*** should be resected (📖 p800). VATS approach is reasonable unless large thymoma or thymic carcinoma which require sternotomy.

Mediastinal compartments

Mediastinal compartments are depicted on 📖 p41.

- ***Anterior superior:*** borders are the thoracic inlet superiorly, sternum anteriorly, pleura laterally, and anterior pericardium posteriorly. Contents include the thymus, lymphatics, and great vessels.
- ***Middle:*** borders are the anterior and posterior pericardium, and pleura laterally. Contents include the heart, phrenic nerves, tracheal carina and bronchi, and lymphatics.
- ***Posterior:*** borders are posterior pericardium anteriorly, spine posteriorly, and pleura laterally. Contents include esophagus, vagi, sympathetic chain, thoracic duct, descending aorta, azygos, and hemiazygos.

Pathology

- ***Etiology***: neurogenic (20%), thymomas (20%), primary cysts (20%), lymphomas (13%), germ cell tumors (10%).
- ***Location***: anterosuperior (54%), posterior (26%), middle (20%).
- ***Malignancy***: 25–50% are malignant. Mediastinal malignancy has highest incidence between the first and fourth decades of life. Malignant neurogenic tumors and non-Hodgkin's lymphoma are the most common mediastinal tumors in children.

Presentation and investigation

1/3 of patients are asymptomatic. Symptoms include chest pain, cough, fever, mechanical compression, and invasion of vital structures. Paraneoplastic syndromes (📖 p670) may be present.

- ***CXR***: provides relative location, incidental discovery.
- ***Chest CT***: defines anatomy and distal spread, bony involvement.
- ***MRI***: able to determine spinal involvement and vascular invasion.
- ***Echo***: evaluation of cardiac tumors and cysts.
- ***Fine needle aspiration or core biopsy***: able to make diagnosis of malignancy in 80–90% of cases. FNA is not reliable for the diagnosis of lymphoma, as additional tissue is needed for adequate flow cytometry.
- ***Mediastinoscopy or anterior mediastinotomy*** (Chamberlain procedure) (📖 p662). Mediastinoscopy is helpful for paratracheal lymphadenopathy (middle compartment) but cannot reach an anterior mediastinal mass. Chamberlain procedure for large anterior mediastinal masses.

Middle mediastinal masses

- ***Pathology***: lymphoma, paratracheal lymph nodes (LNs), bronchogenic cysts.

- ***Lymphoma*** is best diagnosed by mediastinoscopy.
- Although diagnosis by EBUS has been described, often an FNA does not provide enough tissue for diagnosis.
- Hodgkin's lymphoma may be difficult to diagnosis as lymph tissue can be markedly inflamed with ill-defined histology.
- ***Bronchogenic cysts*** account for 5% of mediastinal masses and the most common mediastinal cyst, usually located in the subcarinal area. The majority are asymptomatic, but can cause atypical pain, and symptoms related to compression. May become infected if it communicated with the tracheobronchial tree. If symptomatic, then surgical resection by thoracotomy or VATS.
- ***Thoracic duct cysts***. Very rare. Can present with symptoms of tracheal or esophageal compression. Surgical resection if symptomatic.

Posterior mediastinal masses

- ***Pathology:*** esophageal duplication cysts (see benign disease of the esophagus), neurogenic tumors.
- ***Neurogenic tumors*** make up to 25% of mediastinal masses in adults and 40% in children (most common mediastinal tumor in children). 85% are autonomic ganglia and may produce catecholamines. Occur in the right and left paravertebral sulci, usually in the upper chest, and may involve spinal, intercostal, phrenic, vagus, and sympathetic chain.
- Symptoms include atypical pain, cough, and symptoms of extrinsic compression including dyspnea, dysphagia, and symptoms of neurogenic involvement such as Horner's syndrome, spinal cord compression, also fever and malaise.
- Chest CT and MRI delineate anatomy. MRI especially useful for determining spinal involvement. FNA biopsy.
- Thoracotomy or VATS resection. ***Dumbbell tumors*** or tumors that invade neuroforamina require surgical assistance from orthopedic or neurosurgery, as laminectomy may be required for a R0 resection. Disease-specific adjuvant chemotherapy and/or radiotherapy if malignant.

Thymoma

Key facts

Abnormal persistent growth of the thymus gland instead of the usual involution and atrophy seen in adulthood.

Pathology

- Thymus develops embryonically from the third pharyngeal pouch.
- Lies posterior to the sternal manubrium and anterior to the great vessels, receives its blood supply from the IMAs.
- Venous drainage is to the innominate vein and internal thoracic veins.
- Lymphatic drainage is to the lower cervical, internal mammary, anterior mediastinal, and hilar lymph node lymphatic basins.
- ***Thymoma*** is characterized by thymic epithelial cells with variable lymphocytes, subdivided by fibrous bands.
- ***Thymic carcinoma*** lacks features of normal thymus, characterized by cellular atypia, and increased proliferation of cells.

Presentation

- Usually incidental finding, or paraneoplastic syndromes including:
  - ***Endocrine disorders,*** e.g., Addison's disease, hyperthyroidism, hyperparathyroidism, panhypopituitarism.
  - ***Hematologic disorders,*** e.g., red cell aplasia, hypogammaglobulinemia, T cell deficiency syndrome, pancytopenia, erythrocytosis, amegakaryocytic thrombocytopenia.
  - ***Neuromuscular syndromes,*** e.g., ***myasthenia gravis*** (MG), Eaton–Lambert syndrome, myotonic dystrophy, and myositis.
- 30% of patients with thymoma have MG, 10–20% of patient with MG have thymoma. Thymic carcinoma with MG is rare.
- 5% of patients with thymoma have RBC aplasia and 50% of patients with RBC aplasia have thymoma.
- 2% of patients with thymoma have hypogammaglobulinemia.
- Hypogammaglobulinemia and RBC aplasia indicate poor prognosis.

Classification and staging

Classification includes Muller–Hermelink and World Health Organization (WHO) systems (Table 16.1) and Masaoka clinical stage:

- Stage I: macroscopically completely encapsulated with no invasion.
- Stage II: microscopic capsular invasion or macroscopic invasion into surrounding fatty tissue or mediastinal pleura.
- Stage III: macroscopic invasion into surrounding organs such as the pericardium, great vessels, and pleura.
- Stage IVa: pleural or pericardial dissemination.
- Stage IVb: haematogenous or lymphogenous spread.

Surgery

Main principles include wide resection of entire thymus from phrenic nerve to phrenic nerve, en bloc resection of invaded structure, and marking residual disease with clips for irradiation if unresectable. Preoperative plasmapheresis for thymectomy for MG.

Sternotomy approach

- Upper sternotomy (📖 p467) rather than VATS is approach of choice for larger thymomas or suspicion of thymic carcinoma.
- Open bilateral pleural spaces, divide vein extending from the innominate vein, identify phrenic nerves bilaterally.
- Exenteration of anteromediastinal tissue from phrenic to phrenic nerve, including the pericardial fat, with skeletonization of the innominate vein, and removal intact of all four poles of the thymus.

Transcervical thymectomy

Useful for smaller thymomas, via collar incision in the neck. Surgeon stands at the head of the patient, and a 'thymus retractor' is used to distract sternum anteriorly and expose contents of anterior mediastinum.

VATS thymectomy

- Useful for smaller thymomas, performed from one side, or bilaterally.
- CO_2 insufflation to push the lung aside. May be difficult to identify the contralateral phrenic nerve—bilateral approach limits phrenic injury.
- The patient is placed in the supine position with arms extended, and a shoulder roll along the spine, to allow shoulders to drop posteriorly.
- The pericardial fat pad and thymus is mobilized inferiorly and meticulously moving cephalad and both pleural spaces are opened.
- After freeing the inferior poles, and identifying the contralateral phrenic nerve, the superior poles are mobilized and the thymic vein is divided with a Harmonic scalpel. The specimen is removed in a bag.

Chemotherapy and radiation

- Adjuvant radiation for R1 resection, Stage IIa or higher.
- Adjuvant chemotherapy is platinum based and indicated for R1 resection of thymic carcinoma, R2 resections of thymoma, unresectable disease, or stage III or higher.

Outcomes

- Thymectomy for MG often takes up 1 year to see improvement.
- At 5 years, 30–50% patients will experience some improvement and 25–30% of patients will have complete resolution of symptoms.
- Thymectomy for thymoma, 5-year survival for Masaoka stage—Stage I: 94–100%, Stage II: 86–95%, stage III: 56–69%, stage IV: 11–50.

Table 16.1 Thymoma classification

Muller–Hermelink	WHO
Thymoma	Thymoma
Medullary type	Type A
Mixed type	Type AB
Predominantly cortical	Type B1
Cortical type	Type B2
Well-differentiated carcinoma	Type B3
Thymic carcinoma	Type C (thymic carcinoma)

Anterior mediastinal masses

Key facts

- Masses in the anterior mediastinum are more likely to be malignant (60%) than middle (30%) or posterior (<20%) mediastinal masses.
- Common causes are thymomas (📖 p800), lymphomas (📖 p799) and germ cell tumors, as well as cysts, e.g., pericardial and bronchogenic (📖 p802) which account for around 20% of anterior mediastinal masses.
- Thyroid goiters may have retrosternal extension.
- Thymomas (📖 p800) are the commonest cause of anterior mediastinal masses, although they are rarely seen in children. These should be removed via VATS or sternotomy.

Thymic cysts

- Uncommon in adults. Often found incidentally. May occur in association with syphilis (DuBois abscess) or MG.
- Treatment: thymectomy.

Germ cell tumors

- Same as germ cell tumors in gonads, but are primary and not metastatic. Represent 15–25% of all anterior mediastinal masses.
- More common in young adults and children. 60% are malignant.
- Majority of benign tumors are teratomas.
- Malignant tumors (Table 16.2) include non-seminomas (60%) (embryonal cell, choriocarcinoma, yolk-sac tumors, teratocarcinoma), and seminomas (40%).

Lymphangioma (cystic hygroma)

- Multiloculated cystic mass of lymphatics. Often extends from a cervical or axillary origin. Usually involves the anterior mediastinum.

Table 16.2 Germ cell tumors

α-fetoprotein and β-HCG	Seminomas	Non-seminomas
	Unusual	90%
Associated pathology	None	Chromosomal defects
Metastasis	Unusual	Frequent
Radiosensitive	Very	No
Therapy	Radiation	Platinum-based neoadjuvant chemotherapy; resection if residual disease
Remission	> 80%	Complete response in 55%
		Partial response 35%
Outcomes	5-year survival 50–80%	5-year survival 50–60%

- Chest CT.
- Surgical resection if symptomatic. May be difficult to achieve a R0 resection, especially if it extends into vital structures.

Pericardial cysts

- Most commonly occurs at the right pericardiophrenic angle. Often are found incidentally, but may have symptoms of atypical chest pain. Has no malignant potential.
- Chest CT.
- Surgical resection if symptomatic. VATS resection is the procedure of choice. Care must be taken to avoid injury to the phrenic nerve.

Parathyroid adenoma

- Ectopic anteromediastinal location of a parathyroid adenoma, often found with in a residual thymus gland or area as a result of common embryogenesis from the third branchial cleft.
- Symptoms of primary hyperparathyroidism.
- Sestamibi scan locates tumor in the anterior mediastinum.
- Chest CT confirms location.
- Surgery is VATS resection.
- Preoperative and intraoperative PTH levels are measured to confirm removal.
- The patient can also be injected with sestamibi 1h before surgery, and the lesion can be confirmed after resection with a gamma probe and counter.

Sclerosing fibrosis of the mediastinum

- ***Pathology***: it is a chronic granulomatous proliferative inflammatory process involving the anterior and middle mediastinum. The process can encase mediastinal structures leading to compression-type abnormalities.
- Causes include histoplasmosis, aspergillosis, *Cryptococcus*, TB, nocardiosis, actinomycosis, autoimmune diseases, sarcoidosis.
- More common in females, higher incidence in the midwest USA.
- ***Presentation:*** symptoms include fever, cough, dyspnea, recurrent pneumonias, fistulas of luminal origins, SVC syndrome, cardiac tamponade, cor pulmonale.
- ***Investigation:*** chest CT identifies loss of normal tissue plains, calcified lymph nodes and invasion of adjacent structures.
- ***Management:*** surgical biopsy of lymph nodes for tissue diagnosis.
- Treatment of underlying disease process, including antifungals.
- Alleviation of sequelae, such as repair of SVC syndrome, resection of fibrosed dysfunction tissue, such as pulmonary resection. As tissue planes are lost, may be difficult to identify and control vascular structures. Consider cardiopulmonary bypass for difficult resections.

Thoracic sympathectomy

Key facts

- Interruption of the upper thoracic sympathetic chain (between T2 and T4) usually via bilateral VATS approach: the precise level of interruption remains controversial.
- Indicated for primary hyperhidrosis of the face, and upper extremities and axilla.
- Medical treatment includes topical aluminium chloride, systemic or topical anticholinergic agents, iontophoresis, or localized botulinusm toxin injections. Surgery is reserved for failed medical management.
- Secondary hyperhidrosis, i.e., hyperthyroidism or other systemic entities, should be ruled out before surgical therapy.

Hyperhidrosis

- Most patients report excessive (pools of water rather than just dampness) plantar and palmar sweating since childhood, which has significant impact on work and social life. Commonly associated with axillary hyperhidrosis.
- Incidence around 1–2% depending on definition, and familial history in around 2/3 of patients. It affects females slightly more than males.
- Secondary hyperhidrosis.
- Preganglionic sympathetic fibers responsible for palmar sweating arise from T3–T6, ascending via T2 to reach the hand via the brachial plexus. The sympathetic chain is located just under the rib heads: T2 ganglion is between the heads of the 2nd and 3rd ribs.

Procedure

- Principles of the procedure involve transection of the appropriate level of sympathetic chain that corresponds to the anatomic location of the patient's symptoms, which can be achieved in several ways:
  - ***Sympathectomy*** is removal or ablation of a sympathetic ganglion, as well as transection of the sympathetic chain above and below.
  - ***Sympathicotomy*** is transection of the sympathetic chain.
  - ***Clipping*** refers to application of clips at a particular level, or above and below a specific ganglion.
- The level of division may be based on clinical findings, although multiple trials with conflicting results have been published:
  - For facial sweating the sympathetic chain is isolated by dividing the chain overlying the 2nd and 3rd ribs.
  - For palmar sweating, the chain overlying 2nd, 3rd and 4th ribs is divided.
  - For axillary sweating, the ramus overlying 3rd, 4th, and 5th ribs are divided.
  - Isolated axillary hyperhidrosis is treated by transection of the chain between the 3rd and the 4th rib.
- Care must be taken to avoid injury to the T1 or stellate ganglia, which may lead to Horner's syndrome.

Set-up

- The patient is placed in the semi-Fowler's position (sitting up at about 30–45° to allow the lungs to fall away from the apex), with the arms extended at 90°.
- Two 5mm trocars are placed in the 2nd and 4th intercostal space in the anterior axillary line.
- CO_2 insufflation can be used to further displace the lung (10mmHg).

Sympathectomy

- The sympathetic chain is divided over the appropriate rib with hook cautery (Fig. 16.1). Great care is taken to correctly identify the 2nd rib:
  - It is usually the most proximal rib that can be seen in the thorax.
  - It is the only rib crossed lateral to the sympathetic chain by a small arterial branch of the subclavian that forms the 2nd intercostal artery.
- Both sides are done during the same operative setting.
- The accessory nerve of Kuntz is divided, cauterizing the parietal pleura running along the rib lateral from the rib head.

Outcomes

- Improvement or resolution of symptoms occurs in >90% of patients.
- Compensatory symptoms occur in up to 40% of patients.
- Compensatory sweating includes Frey syndrome or gustatory sweating, sweating of the back, chest, or feet.
- Treatment of compensatory symptoms is generally supportive.
- Asians may have a risk of compensatory sweating of up to 90%.
- Horner's syndrome (ptosis, miosis, anhidrosis) occurs in <1%.
- Resting and peak heart rate decreases by 5–10% as a result of division of sympathetic cardiac nerve supply: this is the rationale for use of sympathectomy in refractory tachyarrhythmias.

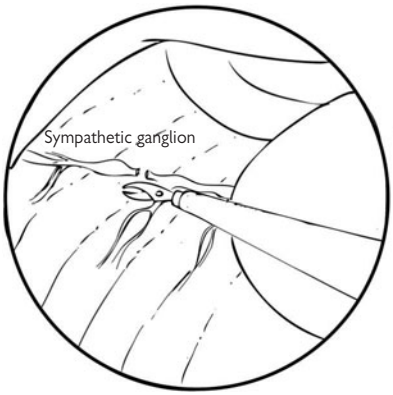


Fig. 16.1 Sympathectomy. The cut is ideally made on the surface of the rib, not between them.

Primary chest wall tumors

Key facts

Primary chest wall tumors comprise 5% of all thoracic neoplasms, and approximately 26% of all chest wall tumors, with 50% being malignant.

Primary bone tumors

- ***Pathology***: 85% occur in the ribs and 15% in the sternum, 90% are malignant. Most common benign tumors are chondroma, fibrous dysplasia, osteochondroma (see Table 16.3).
- Most common malignant tumors are chondrosarcoma, Ewing sarcoma, osteosarcoma, benign primary bone tumors.

Chondroma

- 15% of benign rib tumors, occurs in the second and third decades of life, found in the costochondral junction, difficult to distinguish from chondrosarcoma, histology demonstrates lobules of hyaline cartilage.
- Managed as a malignant lesion (see Chest wall resection, 📖 p808).

Fibrous dysplasia

- Fibrous replacement of the medullary cavity of the rib, slow growing, and generally asymptomatic. Occurs equally in males and females.
- Characteristic radiographic appearance is an expansile lesion with cortical thinning and a ground-glass center.

Osteochondroma

Occurs at the metaphyseal area of the rib, bony stalk with cartilaginous cap, occurs in children. Has ability to transform into a malignancy.

Malignant bone tumors

Chondrosarcoma

- Represents 1/3 of primary malignant bone tumors, and is the most common. Occurs at the costochondral junction of the ribs and sternum. Occurs in the third and fourth decades of life, with a predilection towards males. May have a relationship with chest wall trauma. Grows slowly, with symptoms of pain.
- Treatment is complete resection, 65% 5-year survival, predictors of survival are R0 resection and grade of tumor.

Ewing sarcoma

- 15% of malignant primary chest wall tumor. Occurs in the flat bones and midshaft of long bones, 25% of patients have metastases. 70% of patients develop metastases during up-front therapy. Most commonly occurs in the second decade of life, with predilection towards males.
- Symptoms include pain, enlarging mass, fever, leukocytosis, and ↑ESR.
- Treatment involves upfront chemotherapy, followed by external beam radiation or resection if residual tumor. 60–70% 5-year survival.

Osteosarcoma

- 10% of malignant primary chest wall tumors.
- 1/3 of patients have metastases. Treatment is complete resection.
- Associated with previous radiotherapy, chemotherapy, Paget disease.

- Painful rapidly enlarging mass with elevated alkaline phosphatase. Often in patients age 10–25 and >40 years. 15% 5-year survival.

Plasmacytoma

- Located in the ribs. Occurs in sixth decade of life, commoner in males.
- Treatment is radiotherapy, followed by surgical excision if radioresistant, and chemotherapy for disease progression. 60% 5-year survival.

Primary soft-tissue tumors

- Common benign tumors include connective tissue tumors, e.g., fibroma, lipomas, hemangiomas. Malignant degeneration is rare.
- Treatment is local excision.

Desmoid tissues

- Though technically benign, behaves in an aggressive 'malignant' manner. Originate in the muscle and fascia, and grows along tissue planes and surrounds local structures. histology demonstrates fibroblasts and fibrocytes. Associated with Gardner syndrome.
- Treatment is wide complete resection. High recurrence rate.

Sarcoma

- 50% of all primary malignant chest wall tumors, majority of secondary or metastatic chest wall tumors. 2:1 male to female ratio. Most common in adults, though rhabdomyosarcoma occurs more commonly in children. Most common sarcomas are fibrosarcoma, liposarcoma, rhabdomyosarcoma. Prognostic factors include tumor grade, presence of metastases, and symptoms of pain.
- Treatment is wide complete resection, 82% 5-year survival with low grade tumors compared to 39% with high grade.
- Adjuvant chemotherapy 9 radiation should be considered.

Table 16.3 Primary tumors of the chest wall

	Bone and cartilage	Soft tissue
Benign	Aneurysmal bone cyst	Fibroma
	Chondroblastoma	Haemangioma
	Chondroma	Lipoma
	Giant cell tumor	Lymphangioma
	Eosinophilic granuloma	Neurofibroma
	Fibrous dysplasia	
	Osteoblastoma	
	Osteochondroma	
Malignant	Condrosarcoma	Desmoid
	Ewing's sarcoma	Leiomyosarcoma
	Lymphoma	Liposarcoma
	Osteosarcoma	Lymphangiosarcoma
	Solitary plasmacytoma	Malignant fibrous histiocytoma

Secondary chest wall tumors

Key facts

These include tumors of direct invasion such as breast and lung and metastatic tumors.

Directly invasive secondary chest wall tumors

Breast cancer

- Treatment is resection *if* locoregional disease is controlled, *and* the chest wall is the only site of recurrence *and* R0 resection is achievable.
- 34–58% 5-year survival.

Lung cancer

- Chest wall invasion is considered a T3 tumor. Occurs in 2–8% of patients with NSCLC. Treatment is en bloc resection with 2cm margins.
- 25–40% 5-year survival.

Metastatic secondary tumors of the chest wall

- ***Pathology:*** sarcomas and carcinoma are most common: 20% of all chest wall tumors. Resection *if* locoregional disease is controlled, *and* chest wall is the only metastatic site, *and* R0 resection achievable.
- ***Presentation:*** generally asymptomatic, pain as the lesion becomes larger: painful masses are generally malignant. Bone and cartilage tumors have pain as a common symptom. Systemic symptoms include fever, leukocystosis, eosinophilia, elevated ESR.
- CT: imaging of choice, provides anatomy and delineates associated thoracic pathology. MRI not always necessary. Helpful in determining tissue planes.
- Tissue diagnosis: fine needle or core biopsy. 790% diagnostic. Excisional biopsy for tumors <5cm.
- Reoperation with wide en bloc resection if malignant.
- Incisional biopsy for tumors ≥5cm.

Chest wall resection

- Margins should include one rib above and below the lesion.
- 2cm margins for benign or metastatic lesions.
- 4cm margins for primary malignant lesions such sarcoma.
- En bloc resection of adherent structures.
- Frozen section intraoperative histology of soft tissue and bone marrow to confirm R0 resection.

Chest wall reconstruction

- Defects that are <5cm and/or are covered by the scapula generally do not need reconstruction.
- Reconstruction of large defects avoids paradoxical movement of chest wall with respiration, protects vital organs and improves cosmetics.

Reconstruction options

- Prosthetic materials:
  - Meshes, e.g., Gore-Tex, Marlex, Prolene, PTFE, Vicryl.

- Concrete, e.g., methyl methacrylate, a plastic monomer that is used often in orthopedics. Hardens after drying into cement-like substance. Applied wet between layers of mesh to assist in suturing, and then allowed to dry and harden.

Muscle flaps

- *Latissimus dorsi:* arterial pedicles are the thoracodorsal and lumbar/intercostal arteries. Can cover large areas of the anterior or posterior chest wall. May not be usable if divided during a posterolateral thoracotomy.
- *Pectoralis major:* arterial pedicles are the thoracoacromial artery internal mammary artery perforators and the perforators from the lateral thoracic artery. Excellent for lateral and anterior chest wall defects. Often used as a myocutaneous flap.
- *Rectus abdominus:* pedicles include the superior epigastric artery superiorly and the inferior epigastric artery inferiorly. Useful for sternal defects. Also can be used as a myocutaneous flap.
- *Serratus anterior:* pedicle is derived from the subscapular artery at the origin of the thoracodorsal artery. Primarily used or intrathoracic applications as it has a limited arc of rotation. Often spared during previous posterolateral thoracotomy.

Omentum

- Durable 'wonder' flap that can be used to cover many defects: with skilled mobilization, it can reach the top of the skull.
- The pedicle is either the right (for right-sided defects) or left (for left-sided defects) gastroepiploic artery. Prior to dividing either artery a vascular clamp should be placed on the proposed pedicle to insure that it is adequate to supply all arcades of the omentum.

Chest wall deformities

Pectus exavatum ('funnel chest')

Pathology

- Anterior thoracic wall deformity with depression of the sternum and lower costal cartilages. Frequently asymmetrical with the right side more depressed than the left. The manubrium and first two costal cartilages are usually normal.
- 1 in 300–400 live births, 5:1 male to female ratio. 90% noted with the first year of life. Family history in 37% of patients.
- Posterior traction of sternum due to diaphragm abnormalities or excessive pressure in utero. Abnormal production of connective tissue.

Presentation

- Often asymptomatic, presenting for cosmetic reasons. Dyspnea and easy fatigability. Transient atrial arrhythmias manifesting as palpitations.
- Cardiovascular changes: up to 28% drop in CO and 31% drop in SV.
- Pulmonary function changes: decreases in VC, TLC, and MVV.

Indications for surgery

- Cosmetic reasons causing emotional distress, symptoms, high degree of deformity (severity index >3.2). Severe sternal depression.
- Repair after age 8.
- *Modified Ravitch procedure* (see Fig. 16.2): submammary incision. Subperiosteal excision of the affected costal cartilages with elevation of the sternum via a sternal wedge osteotomy. Insertion of a pectus bar to maintain position of the elevated sternum.
- *Nuss procedure* (see Fig. 16.2): a Nuss bar is placed thoracopically with no costal cartilage excision. The bar is rotated, causing elevation of the sternum.
- Complications include wound infection, pain, pneumothorax, migration of the Nuss bar, recurrence and avulsion of pectus muscles.

Pectus carinatum ('pigeon breast')

Pathology

- Anterior chest wall deformity with protrusion of the mid to lower sternum, with lateral depression of the ribs.
- 4:1 male to female ratio. 22% of patients have concurrent scoliosis and 20% congenital heart disease. 25% family history chest wall disorders.

Presentation

Often symptomatic. Dyspnea, with diminished respiratory excursion of the chest. Easy fatiguability and atypical chest pain. Psychological distress from poor cosmetics.

Options for reconstruction

Submammary incision with subpericondral resection of deformed costal cartilages, sternal wedge osteotomy for posterior displacement of the sternum, removal of the xiphoid process.

Complications

Wound infection, recurrence, and pneumothorax.

Poland's syndrome

Pathology

- Congenital absence of the pectoralis muscles. Varying degrees of breast, hand, and anterior chest wall bony defects.
- 1 in 30 000 live births. Unknown etiology.

Surgical options

- Small deformities limited to the absence of the medial part of the pectoralis muscles may not require major reconstruction.
- Extensive deformities require major reconstruction with sternal wedge osteotomy, rib grafts taken from the contralateral side, resection of depressed costal cartilages, insertion of polypropylene mesh, latissimus dorsi flap for soft tissue coverage, and breast augmentation in females.

Other chest wall deformities

- Cleft sternum.
- Thoracic ectopia cordis.
- Rib dysplasia.
- Asphyxiating thoracic dystrophy (Jeune syndrome).
- Spondylothoracic dysplasia (Jarcho–Levin syndrome).

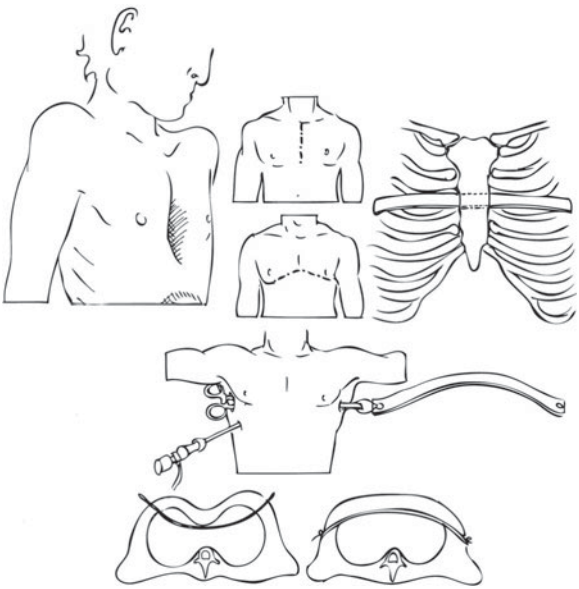


Fig. 16.2 Options for pectus excavatum. Top: modified Ravitch procedure. Bottom: Nuss procedure.

Thoracic outlet syndrome

Key facts

Compression of the brachial plexus and subclavian vessels at the apical opening of the thorax (thoracic outlet).

Pathology

Congenital narrowing of the thoracic outlet may be caused by:

- Congenital lesions: cervical rib, first thoracic rib, abnormal anterior scalene muscle, biphid clavicle, enlarged transverse process of the C7 vertebrae, anomalous course of the transverse cervical artery, adventitious fibrous bands.
- Trauma causing narrowing of the thoracic outlet, e.g., clavicular fracture, head of humerus dislocation, crush injury, cervical spondylosis, cervical scar tissue, severe atherosclerosis.

Presentation

- ***Neurologic symptoms*** are the most common: pain paresthesias of the ulnar brachial plexus C8 and T1 distribution, motor weakness of the hands, atypical anterior chest wall pain, headaches.
- ***Arterial compression*** symptoms are the next common: poikilothermia, Raynaud's phenomenon in 7.5% of patients, effort fatigue of the arm and hand, diffuse pain.
- ***Venous compression*** symptoms are rarer: effort thrombosis of the subclavian and axillary vein (Paget–Schroetter syndrome).

Physical examination

There are a number of physical tests that make diagnosis of TOS likely:

- *Adson (Scalene) test:* occludes the radial pulse and/or symptoms are reproduced when the head is turned to the ipsilateral side and the patient breathes in.
- *Halsted test:* occludes the radial pulse and/or symptoms are reproduced when the shoulders are displaced caudally (military position).
- *Wright test:* occludes radial pulse and/or symptoms are reproduced when the ipsilateral arm is abducted cranially and the elbow is extended, and arm is elevated for 1min. Symptoms may occur sooner if digital pressure is placed in the ipsilateral supraclavicular fossa.
- *Roo's test:* occludes the radial pulse and/or symptoms are reproduced with 3min of rapid opening and closing of the ipsilateral hand.

Investigations

The diagnosis of TOS is often a diagnosis of exclusion:

- Differential diagnosis includes carpal tunnel syndrome, cervical disc prolapse, superior sulcus tumors, collagen vascular disorders, Raynaud disease.
- Chest and cervical plane films may reveal cervical rib or prolonged transverse processes.
- Cervical and thoracic cervical CT may identify cervical and thoracic pathology. Commonly, imaging is unremarkable.
- Sensory and nerve conduction studies are often unremarkable, but may be helpful in ruling out other causes.

Management

- Physical therapy is the first-line treatment.
- Surgery is reserved for patients who fail physical therapy and involves (1) resection of the first rib, (2) vascular and brachial plexus decompression, (3) division of costoclavicular ligament, (4) resection of the anterior and middle scalene muscles, and (5) sympathectomy C8 through T3.

Transaxillary approach

- This is preferred for symptoms of nerve compression: as it allows the first rib and soft tissues to be completely resected.
- The patient is placed in the lateral decubitus position.
- Transaxillary incision just below the hairline between the pectoralis major and latissimus dorsi.
- The assistant should elevate the upper arm towards the ceiling to open up the axilla.
- The insertion of the anterior scalene muscle is divided where it is attached to the first rib.
- A middle wedge of the first rib is resected with bone cutters.
- Anterior rib is resected to the costocartilage of the sternum and the posterior rib is resected to the transverse process.
- A T1–T3 sympathectomy is optional.

Supraclavicular approach

- This is used for reoperations (the posterior approach shown in Fig. 16.3) is an alternative for reoperations.
- A supraclavicular cervical incision is made parallel to the clavicle.
- The scalene fat pad is removed to identify the phrenic nerve running anterior to the anterior scalene muscle.
- The anterior scalene is divided revealing the subclavian artery.
- The brachial plexus is retracted to reveal the first rib.
- The first rib is removed.

Complications

- 10% recurrence rate.
- Resection of the wrong rib (second rib), pain, injury to vessels or nerves.



Fig. 16.3 Transaxillary approach to first rib resection.

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